Power BI Interview Question Bank

Scenarios, DAX & advanced solutions · CrackAnalytics · by Mahendra Singh

Explore more — click on this link: https://medium.com/@mahendraa1188

Scenario-based questions (full answers)

Q1. What did you do with your project and what are your roles & responsibilities? [Easy]

In my project, I worked on building an interactive Power BI dashboard for a retail business to analyze sales performance, customer behavior, and product demand. My key responsibilities included:

Data Extraction & Transformation: Collected data from multiple sources like SQL databases, Excel, and cloud storage. Cleaned and transformed the data using Power Query.

Data Modeling: Created relationships between Fact and Dimension tables and implemented Star Schema for optimized performance.

DAX Calculations: Developed complex DAX measures for KPIs like Total Sales, Profit Margins, and Year-over-Year Growth.

Report & Dashboard Design: Designed visually appealing and user-friendly dashboards with slicers, drill-throughs, and interactive visuals.

Performance Optimization: Used Import Mode for better speed and optimized DAX queries for efficiency.

Collaboration & Deployment: Published reports to Power BI Service, scheduled data refreshes, and shared insights with stakeholders.

Q2. What are the transformations used in your project? [Easy]

Removing Duplicates: Ensured unique records for accuracy.

Changing Data Types: Converted columns into appropriate formats (e.g., text to date, numbers to currency).

Splitting Columns: Used Split Column for separating full names into first and last names.

Merging Queries: Combined multiple datasets for a unified view.

Adding Custom Columns: Created calculated columns like Profit = Sales – Cost.

Unpivoting Data: Converted wide-format data into a normalized structure for better analysis.

Example: While working on a sales dataset, I had to split the "Date Time" column into separate "Date" and "Time" fields for better filtering and analysis.

Q3. What are the different sources you have used in your project? [Easy]

I have worked with multiple data sources, including:

- **SQL Server / MySQL / PostgreSQL / Snowflake:** Extracting structured data using SQL queries.
- **Excel / CSV Files:** Handling offline data.
- **SharePoint / OneDrive:** Importing data stored on cloud platforms.
- **REST APIs:** Fetching real-time data from web services.
- **Google Sheets:** Integrating live data from Google Workspace.
- **Azure / AWS / Google Cloud:** Connecting to cloud databases and data lakes.

Example: In a financial project, I connected Power BI to an Azure SQL database and merged it with Excel-based budgeting data to create an interactive expense tracker.

Or you can create your own story based on your recent project.

Q4. You have sales data from multiple regions stored in different tables (Sales, Products, Customers). How would you design a data model in Power BI for efficient reporting? [Medium]

- Use a star schema design with a central fact table (Sales) and dimension tables (Products, Customers, Regions).
- Create relationships between the fact table and dimension tables using primary/foreign keys (e.g., Product ID, Customer ID).
- Ensure relationships are single-directional and avoid circular dependencies.
- Use DAX to create calculated columns or measures (e.g., Total Sales = SUM(Sales[Amount])).

Star Schema Implementation:

1. Fact Table (Sales):

- Contains all transactional data (sales records)
- Includes foreign keys to dimension tables
- Stores quantitative measures (sales amount, quantity, profit)

2. Dimension Tables:

- **Products:** Product ID, Name, Category, Subcategory, Price
- **Customers:** Customer ID, Name, Segment, Region, Contact
- **Dates:** Date ID, Full Date, Day, Month, Quarter, Year
- **Regions:** Region ID, Country, State, City, Postal Code

A star schema simplifies data modeling and improves query performance. Power BI's engine is optimized for this structure, enabling faster aggregations and filtering.

Q5. Difference between Star Schema and Snowflake Schema? *[Easy]*

- Star schema | Snowflake schema
- Dimensions are denormalized — one table per dimension | Dimensions are normalized into sub-tables (Product → Category → Subcategory)
- Fewer joins, faster query performance | More joins, slightly slower queries
- Simpler to understand and maintain | Saves storage, but more complex
- Preferred structure in Power BI | Used when dimension tables are very large

Q6. Your Power BI report is slow. What steps would you take to optimize its performance? *[Hard]*

1. Data Model Optimization

Star Schema Design

- Central fact table (e.g., Sales) linked to dimension tables (e.g., Products, Customers).
- Ensures efficient filtering and reduces storage overhead.

Column & Row Reduction

- Remove unused columns (especially high-cardinality text fields).
- Filter out unnecessary historical data (e.g., keep only the last 3 years).

Relationship Optimization

- Use **integer keys** (not text) for joins (e.g., Product ID instead of Product Name).
- Set **single-directional** cross-filtering (dimension → fact).
- Avoid bi-directional relationships unless absolutely necessary.

2. DAX & Calculation Optimization

- **Measures > Calculated Columns** — measures compute at query time (dynamic); calculated columns consume memory.
- Replace `SUMX(` with `SUM(` when possible for better performance.
- **Optimize time intelligence:** use `TOTALYTD`/`DATESYTD` instead of manual date filtering; pre-calculate rolling metrics (e.g., Rolling 12M Sales) in the data source if possible.
- **Avoid expensive functions:** replace nested `CALCULATE` with `SUMMARIZE` or `ADDCOLUMNS`; use `DISTINCTCOUNT` sparingly — consider pre-aggregating in SQL.

3. Data Source Optimization

- **Query Folding (Power Query):** ensure transformations (filters, joins) push back to the source (SQL, etc.). Use **View Native Query** to verify folding.
- **Storage Mode Selection:** Import Mode — best for small–mid datasets (fastest in-memory queries). DirectQuery — for large/real-time data (but slower visuals). Hybrid — aggregate tables in Import + details in DirectQuery.
- **Incremental Refresh:** for large fact tables, refresh only new data (e.g., WHERE Order Date >= TODAY() – 30).

4. Report-Level Optimization

- **Visual & page limits:** max 5–10 visuals per page (each visual runs separate queries). Use bookmarks or drill-throughs instead of overcrowding.

- **Filter efficiency:** apply page-level filters before visual-level filters; use slicers with "Single select" to reduce DAX overhead.
- **Other tips:** disable interactions between non-linked visuals; use static images/icons instead of shape visuals.

5. Advanced Techniques

- **Aggregation tables:** pre-summarize data (e.g., daily sales by region) for faster queries.
- **Calculation groups (Tabular Editor):** reuse measure logic (e.g., MTD/QTD/YTD) without DAX duplication.
- **Performance Analyzer:** use View → Performance Analyzer to identify slow visuals/DAX.

Optimizing the data model, DAX, and report design ensures faster load times and better user experience.

Q7. You have a large dataset that updates daily. How would you implement incremental refresh in Power BI? [Hard]

- In Power Query, partition the data using a date column (e.g. Order Date).
- In Power BI Desktop, enable Incremental Refresh in the dataset settings.
- Set parameters for `RangeStart` and `RangeEnd` to define the refresh window.
- Configure the incremental refresh policy (e.g., keep 2 years of historical data and refresh the last 7 days daily).

Incremental refresh reduces the amount of data processed during each refresh, improving performance and reducing resource consumption.

Q8. A client wants to see the distribution of their customer base by age group and purchasing behaviour. How would you create a segmentation analysis with interactive filtering? [Hard]

- Create an Age Band calculated column (e.g. 18–25, 26–35, 36–50, 50+) or a separate segmentation table.
- Build measures for purchasing behaviour — purchase frequency, average order value, total spend.
- Use a scatter chart (spend vs frequency, coloured by age band), bar charts by segment, and slicers for interactive filtering.
- Add drill-through to a customer-detail page for deeper insights on any segment.

Q9. Data Modeling Case: Sales data and customer data are in separate tables. How would you model this to analyze customer purchase behaviour? [Medium]

Load the Data: Import the sales data and customer data tables into Power BI. Establish relationships: identify the CustomerID as the common key between the two tables. In the "Model" view, create a relationship by connecting the CustomerID column from the Sales Data table to the CustomerID column in the Customer Data table.

Data Structure: Sales Data Table contains columns like SaleID, CustomerID, ProductID, SaleDate, and Amount. Customer Data Table contains columns like CustomerID, CustomerName, Age, Gender, and Location.

Create Visualizations:

- **Total Sales by Customer:** a bar chart showing the total amount spent by each customer.
- **Sales Over Time:** a line chart displaying sales trends over time for each customer.
- **Customer Demographics:** pie charts or bar charts illustrating sales distribution by customer age, gender, and location.

Utilize DAX for Advanced Analysis: create measures using DAX (Data Analysis Expressions) to calculate total sales and sales by specific customer attributes for deeper insights. By following these steps, you can effectively model your data in Power BI to gain meaningful insights into customer purchase behaviour.

Q10. How would you handle a situation where your Power BI report is performing slowly? What steps to diagnose and fix? [Hard]

To handle a situation where a Power BI report is performing slowly, you can:

- Optimize your data model by removing unnecessary columns and tables.
- Use relationships and filtering carefully to minimize the amount of data processed.
- Avoid using complex DAX calculations in visuals; instead, create calculated columns or tables if needed.
- Use aggregate tables or pre-aggregated data to reduce the volume of data processed in visuals.
- Ensure that your data source is optimized for performance, such as indexing important columns or partitioning large tables.
- Use Power BI Performance Analyzer to identify and troubleshoot performance bottlenecks in your report.

Or, step by step:

- **Check Data Volume:** large datasets can slow down your report. Try to reduce the amount of data by filtering or aggregating it.
- **Optimize Data Model:** remove any unnecessary columns or tables. Use appropriate data types and relationships.
- **Review DAX Calculations:** simplify complex DAX formulas. Avoid using too many calculated columns or measures.
- **Manage Visualizations:** limit the number of visuals on a single report page. Use simpler visuals when possible.
- **Reduce Query Load:** use "Query Folding" to push operations back to the data source. Make sure your queries are efficient and optimized.
- **Enable Performance Analyzer:** in Power BI Desktop, go to "View" → "Performance Analyzer." Run it to see which visuals or queries are taking the most time.
- **Optimize Data Refresh:** schedule refreshes during off-peak hours. Ensure incremental refresh is set up if possible.
- **Improve Power BI Service Settings:** ensure your Power BI workspace is in the correct region. Check for any limitations or restrictions on the service.

Q11. There is a report with five visuals and a slicer. If the slicer is changed, only two visuals should be affected; the remaining three should be unaffected. What are you going to do? [Easy]

Select the slicer visual → Go to the "Format" tab and click "Edit interactions" → Set the desired interaction for each visual → Exit the "Edit interactions" mode.

Q12. There are 2 pages in a report. Page 1 has a Country slicer. If country is changed on page 1, page 2 should automatically be impacted too. How will you do it? [Easy]

Select the country slicer → Open the **Sync Slicers** pane from the View tab → Check the Sync boxes for both pages (Page 1 and Page 2).

Q13. Report consists of many visuals and some of the visuals are loading very slowly? [Medium]

Reduce Data Size → Optimize Calculations → Use Star Schema → Limit the Number of Visuals → Performance Analyzer.

Q14. You are a data analyst for a global e-commerce company. You need to analyze marketing campaign performance across regions and identify campaigns with the highest ROI, plus how customer acquisition cost (CAC) varies by region and campaign. How would you build this report? [Hard]

- Bring campaign spend, conversions and revenue data together; model campaigns, regions and dates as dimensions.
- Create DAX measures: `ROI = DIVIDE([Revenue] − [Spend], [Spend])` and `CAC = DIVIDE([Spend], [New Customers])`.
- Build a matrix of campaign × region with ROI and CAC, a map or bar chart for regional comparison, and trend lines over time.
- Add slicers for region/campaign/date and drill-through to campaign-level detail so stakeholders can find the highest-ROI campaigns instantly.

Q15. Imagine you need to visualize year-over-year growth in product sales. What approach would you take to calculate and present this effectively? [Medium]

To visualize year-over-year growth in product sales, I would first calculate the sales for each product for the current year and the previous year using DAX measures in Power BI. Then, I would create a line chart visual where the x-axis represents the months or quarters, and the y-axis represents the sales amount. I would plot two lines on the chart, one for the current year's sales and one for the previous year's sales, allowing stakeholders to easily compare the growth trends over time.

Q16. You're working with a dataset that requires extensive data cleaning and transformation before analysis. Describe your process for cleaning and preparing the data in Power BI. [Medium]

For cleaning and preparing the dataset in Power BI, I would start by identifying and addressing missing or duplicate values, outliers, and inconsistencies in data formats. I would use Power Query Editor to perform data cleaning operations such as removing null values, renaming columns, and applying transformations like data type conversion and standardization. Additionally, I would create calculated columns or measures as needed to derive new insights from the cleaned data.

Q17. Your organization wants to incorporate real-time data updates into their Power BI reports. How would you set up and manage live data connections? [Hard]

To incorporate real-time data updates into Power BI reports, I would utilize Power BI's streaming datasets feature. I would set up a data streaming connection to the source system, such as a database or API, and configure the dataset to receive real-time data updates at specified intervals. Then, I would design reports and visuals based on the streaming dataset, enabling stakeholders to view and analyze the latest data as it is updated in real-time.

Q18. How do you work with large datasets in Power BI? [Hard]

When dealing with large datasets in Power BI, the primary challenge is the size of the data, which can affect performance, making the report slow to load and refresh. Managing and visualizing such a vast amount of data requires efficient handling to avoid timeouts and performance degradation.

One of the strategies I use is to upload a subset of the data into Power BI Desktop initially. For example, if I have data spanning five years, I might start by uploading only six months of data. This speeds up the development process on the desktop.

Next, I use the Power Query Editor to filter and aggregate data. This includes removing unnecessary columns, filtering rows to include only relevant data, and aggregating data at a higher level. For instance, if detailed transaction data is not necessary, I might aggregate daily sales data to monthly sales data before loading it into Power BI.

For extremely large datasets, I use DirectQuery mode, which allows Power BI to directly query the underlying data source without importing the data into the Power BI model. This keeps the Power BI model lightweight and leverages the processing power of the database server. However, this requires a well-optimized database and efficient query performance at the source. Sometimes, I use a combination of Import and DirectQuery modes, known as composite models. This approach allows for flexibility by importing critical, smaller tables into the Power BI model and using DirectQuery for larger fact tables.

I ensure that the data model is optimized by creating appropriate relationships and using measures efficiently. Reducing the complexity of DAX calculations and ensuring that the model only includes necessary tables and relationships helps maintain performance.

By employing these strategies, I can manage large datasets efficiently, ensuring that my Power BI reports are responsive and performant.

Advanced DAX & solution bank

Q1. How would you write a DAX formula to calculate a running total that resets every year?

RunningTotal =

CALCULATE(SUM('Sales'[Amount]),

FILTER(ALL('Sales'),

'Sales'[Year] = EARLIER('Sales'[Year]) &&

'Sales'[Date] <= EARLIER('Sales'[Date])))

Q2. How would you manage and optimize Power BI reports that need to handle very large datasets (millions of rows)?

1. Use DirectQuery mode if real-time data is needed.
2. Pre-aggregate data in the data source.
3. Use dataflows for preprocessing.
4. Implement incremental refresh.

Q3. What steps would you take if a scheduled data refresh in Power BI fails?

Check the Power BI service for error messages.

Verify data source connectivity and credentials.

Review gateway configuration.

Optimize and simplify the query.

Q4. How would you create a report that dynamically updates based on user input or selections?

Use slicers and what-if parameters. Create dynamic measures using DAX that respond to user selections.

Q5. How would you incorporate advanced analytics or machine learning models into Power BI?

Use R or Python scripts in Power BI to apply advanced analytics.

Integrate with Azure Machine Learning to embed predictive models.

Use AI visuals like Key Influencers or Decomposition Tree.

Q6. How would you integrate Power BI with other Microsoft services like SharePoint, Teams, or PowerApps?

Embed Power BI reports in SharePoint Online and Microsoft Teams. Use PowerApps to create custom forms that interact with Power BI data. Automate workflows with Power Automate.

Q7. How to use if Parameters in Power BI?

Go to "Manage Parameters":
Navigate to the "Home" tab in the ribbon.
Click on "Manage Parameters" from the "External Tools" group.
Click on "New Parameter."
Enter a name for the parameter and select its data type (e.g., Text, Decimal Number, Integer, Date/Time).
Optionally, set the default value and any available values (for dropdown selection).

Q8. What is the role of Power BI Paginated Reports and when are they used?

Power BI Paginated Reports (formerly SQL Server Reporting Services or SSRS) are used for pixel-perfect, printable, and paginated reports. They are typically used for operational and transactional reporting scenarios where precise formatting and layout control are required, such as invoices, statements, or regulatory reports.

Q9. What are the options available for managing query parameters in Power Query Editor?

Power Query Editor allows users to define and manage query parameters to dynamically control data loading and transformation. Parameters can be created from values in the data source, entered manually, or generated from expressions, providing flexibility and reusability in query design.
Power BI interview questions and answers
10. What is Power BI?
Answer: Power BI is a business analytics service by Microsoft that provides interactive visualizations and business intelligence capabilities with an interface simple enough for end-users to create their reports and dashboards.
11. Differentiate between Power BI Desktop, Power BI Service, and Power BI
Mobile.
Answer: Power BI Desktop is used for creating reports, Power BI Service (or Power BI Online) is the cloud service for sharing and collaborating on reports, and Power BI Mobile allows users to access reports on mobile devices.
12. Explain the role of Power Query in Power BI.
Answer: Power Query is used for data transformation and shaping. It allows users to connect to various data sources, clean and transform data before loading it into Power BI for analysis.
13. What is DAX in Power BI, and why is it important?
Answer: DAX (Data Analysis Expressions) is a formula language used for creating custom calculations in Power BI. It is important as it enables users to create sophisticated measures and calculated columns.
14. What is the difference between a calculated column and a measure in Power BI?
Answer: A calculated column is a column added to a table, computed row by row, while a measure is a formula applied to a set of data, providing a dynamic calculation based on the context.
15. How can you implement row-level security in Power BI?
Answer: Row-level security in Power BI can be implemented by creating roles in Power BI Desktop and defining filters at the row level based on user roles.
16. Explain the purpose of the Power BI Gateway.
Answer: The Power BI Gateway allows for a secure connection between Power BI services and on-

premises data sources. It facilitates refreshing datasets and running scheduled refreshes.

17. What is a Power BI dashboard?

Answer: A Power BI dashboard is a single-page, interactive view of your data that provides a consolidated and visualized summary of key metrics. It can include visuals, images, and live data.

18. How can you share a Power BI